

Chapter 12: The Muscular System and Digestive System in Human Beings

Textbook Exercise Solutions & Answers



1. Fill in the blanks with the right word from the brackets:

(a) The process of digestion starts from the (stomach, mouth).

mouth

(b) Eyelids have muscles. (voluntary, involuntary)

involuntary

(c) is not a function of muscular system. (production of blood cell, performing movement)

Production of blood cells

(d) Muscles of the heart are (ordinary muscles, cardiac muscles)

cardiac muscles

(e) Pushing forward the food that has been chewed is the function of the (stomach, oesophagus)

Oesophagus.

2. Find a match for me.

Column 'A'	Column B'
1. Cardiac muscle	a. Always functions in pairs
2. Are brought about by muscle	b. We never feel tired
3. Pepsin	c. Uncontrolled and painful contraction of muscles
4. Cramps	d. Chewing movement of the jaw
5. Skeletal muscles	e. Enzymes of the gastric juice

3. Who is telling a lie?

Organ Statement 1. Tongue a. My taste buds can tell only a sweet taste. 2. Liver b. I am the largest gland in the body. 3. Large intestine c. I am 7.5 metre long. 4. Appendix d. Digestion is impossible without me 5. Lung e. I play an important role in excretion.

1. Lie. My taste buds can tell all tastes – sweet sour, bitter.
2. Truth.
3. Lie. It is 1.5 metre long.
4. Truth.
5. Lung → Lie. It plays important role in breathing.

4. Give reasons.

(a) Food becomes acidic in the stomach.

1. The gastric glands of stomach secrete gastric juice.
2. Food that has entered stomach is churned.
3. Three components of gastric juice namely hydrochloric acid, pepsin and mucus are mixed with food here and the food becomes acidic.

(b) Cardiac muscles are said to be involuntary muscles.

1. Cardiac muscles are found in the heart.
2. These muscles bring about the contraction and relaxation (beating) of the heart.
3. Their movement is involuntary.
4. Cardiac muscles cause our heart to relax and contract continuously at a rate of about 70 times per minute. They do not depend upon our will.
5. Beating is carried out in their own fixed manner.

So cardiac muscles are said to be involuntary muscles.

(c) Intoxicating substances should not be consumed.

1. Physical health is important for our organ system to function properly.
2. But harmful habits like smoking, chewing of tobacco, drinking alcohol affect our health adversely.
3. If we consume any tobacco products, the mouth, pharynx, alimentary canal, and other organs of the digestive system cannot function properly.
4. It causes problems like vomiting, nausea, and headache.
5. Tobacco particles stick to teeth, gums, and skin of the mouth cavity and slowly cause injury to those parts resulting in their dysfunction.
6. This causes swelling of the gums and pain when moving the jaws.
7. The pharynx and intestine become inflamed it progress into cancer leading to death.

(d) Your muscles should be strong and efficient.

Muscles are bundles of fibres that can contract and relax as required.

1. The action of muscles is necessary for all kinds of movements from the small movements of eyelid to those that demand great strength when chopping wood with an axe.
2. We use muscles for various movements like talking laughing, walking, jumping, throwing etc.
3. Therefore our muscles should be strong and efficient to do our day today work well and smooth functioning of life processes.

5. Answer the following.

(a) How many types of muscles are there? Which are those types?

Muscles are bundles of fibres that can contract and relax as required. There are three types of muscles.

- Skeletal muscles.
- Heart or cardiac muscles.
- Smooth muscles.

1. Skeletal muscles: (a) Skeletal muscles work with bones, the two ends of each of these muscles are attached to two different bones. (b) They are responsible for holding the bones of the skeleton together and giving shape to our body, (c) Skeletal muscles permits movement of the body and maintain the posture of the body. (d) Skeletal muscle is voluntary e.g. muscles in our arms and legs are voluntary muscles, their action depends upon our will. That's why they are called voluntary muscles.

2. Heart or cardiac muscles: (a) Heart or cardiac muscles bring about the contraction and relaxation (beating) of the heart. (b) Their movement is involuntary, (c) Cardiac muscles cause our heart to relax and contract continuously at a rate of about 70 times per minute, (d) Cardiac muscle is found in heart.

3. Smooth muscles: (a) These muscles are present in the internal organs other than the heart, e.g. muscles of the stomach, intestine, blood vessels, uterus etc. (b) Their movements are involuntary and slow, (c) They are not according to our will. (d) Various vital functions of our body such as digestion, respiration and movement of food material of which we remain quite unaware, are carried out by these special muscles.

(b) What causes the problem of acidity? What is its effect on the body?

1. Stress is the main cause of acidity.
2. Hectic lifestyle and stress can lead to unhealthy or irregular meals, not good for digestion process, and this may cause acidity.
3. Other reasons are eating spicy food, drinking too much alcohol, missing meals, an empty stomach may lead to acidity.
4. Acidity leads to stomach upset, burning sensation in chest and stomach, constriction of blood vessels, weight gain, obesity, cardiovascular damage.

(c) Name the different types of teeth. What is the function of each type?

There are four types of teeth, namely incisors, canines, pre-molars, and molars.

Each tooth is covered by a hard substance called enamel. Enamel is made up of calcium salt.

The process of digestion begins with the function of the teeth in the mouth.

1. Incisors: (a) These come in first 6-months of age. Incisors are the eight teeth in the front and centre four on top and four on bottom, (b) They are sharp and blade like for cutting food. e.g. for biting an apple, (c) We use them to take first bite of food.

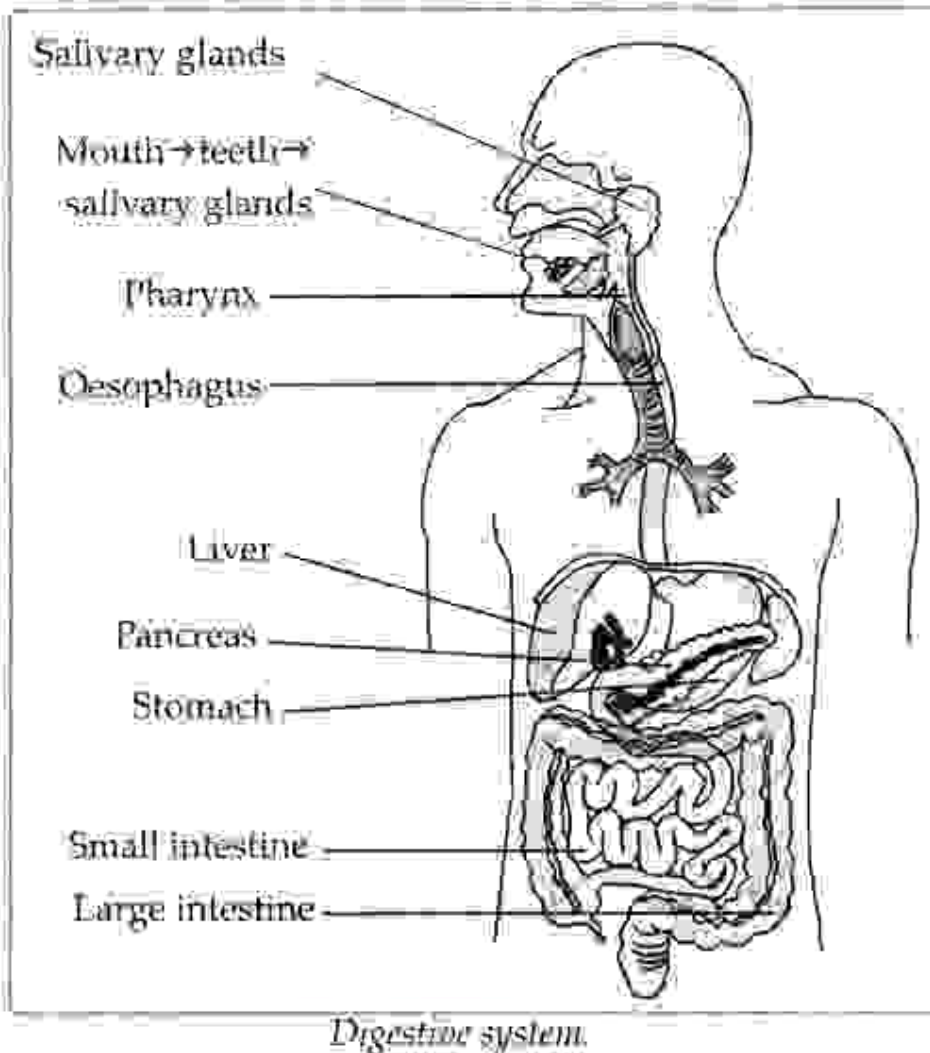
2. Canines: (a) These are strong and pointed, sharpest of all for tearing food. e.g. to tear off a piece of tough meat, (b) They play important role in digestion of food, (c) They appear between 11 and 20 months of age.

3. Pre-molar: Pre molar share features of both canines and molars. (a) You can use them for grinding and chewing food. So that it becomes semi-liquid helping to gulp down the throat easily, (b) They are situated at each side of your mouth in deep settings, (c) They appear at the age of 10 years.

4. Molar: (a) These are broad and flat on top for crushing and grinding food. e.g. to grind up nutmeats. (b) Two teeth above and two teeth below, they appear at the age 11-13 years. (c) Molars are more prone to germ attack because of their remote location in our mouth.

So we should keep them clean.

6. Sketch and label a diagram of the digestive system and describe it in your own words.



1. Conversion of food into a soluble form and its absorption into the blood is called digestion.
 2. The digestive system consists of the alimentary canal and digestive glands.
 3. The total length of alimentary canal is about 9 metres.
 4. Its main parts are the mouth, pharynx, oesophagus, stomach, small intestine, large intestine, rectum and anus.
 5. The salivary glands, liver, pancreas are the digestive glands connected to the alimentary canal.
 6. Different organs of the digestive system perform the function of digestion.
 7. There are different stages in the process of digestion of food.
 8. The process of digestion begins with the function of the teeth in the mouth, food is chewed into small pieces.
 9. There are four types of teeth, incisors, canines, pre-molars and molars.
 10. Saliva in the mouth contains enzymes, ptyalin, or amylase. It converts starch into maltose.
- a. Oesophagus:

1. It is a tube leading from the pharynx to the stomach,
2. It pushes the food towards the stomach.

b. Stomach:

1. The large sac like part of the alimentary canal is called the stomach.
2. Food that has entered is churned.
3. The gastric glands of stomach secrete gastric juice which contain hydrochloric acid, pepsin and mucus.
4. They mix with food and food becomes acidic.
5. Mainly proteins are digested in stomach.
6. Due to the churning and actions of gastric juice, food become a semi-solid slurry which is pushed into the small intestine.

c. Small intestine: (6m long)

1. Bile secreted by liver mixes with food in small intestine,
2. Most of the digestion and absorption of food takes place here.

d. Large intestine: (1.5m long) only water is absorbed in the large intestine. Undigested remain is thrown out of the body through the anus.